



ARCHANA THARAMMAL

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SUMMARY

As an AI/ML Engineer with an MSc in Data Science from Heriot-Watt University, I specialize in designing and deploying machine learning models and data-driven systems to solve complex real-world problems. My expertise spans predictive modeling, feature engineering, model optimization, and LLM-based agents. I build end-to-end AI solutions, including conversational AI systems and real-time ML pipelines, leveraging automation workflows and vector databases to deliver scalable, low-latency, and production-ready systems that generate actionable insights.

EXPERIENCE

AI Solutions Engineer Intern – Gyanwave Technology, DWTC, Dubai

Sept 2025 – Present

RAG-Powered Multi-Channel Systems: Developed a web crawler using Crawl4AI to extract structured and unstructured content from client websites and store it into a Qdrant vector database. Integrated the crawler with RAG pipelines to power multilingual chatbots across web, WhatsApp, and Telegram. Designed a user-friendly production ready UI where clients can paste their company website link to crawl and automatically generate a live chatbot within minutes. This interface allowed clients to preview the chatbot experience and easily integrate it into their websites or other platforms, improving UX evaluation and reducing manual data lookup.

Voice AI Systems & CRM + Telephony Integrations : Designed and developed a full-stack AI voice assistant platform using React, TypeScript, and Vite with enterprise-grade UI components for a responsive frontend experience. Built scalable backend services using Node.js and Express with PostgreSQL, enabling structured storage of conversations, user sessions, and CRM-linked interactions. Integrated AI voice capabilities using OpenAI and ElevenLabs to generate natural, real-time conversational voice responses for customer interactions. Implemented telephony integration using Twilio and Plivo to support inbound and outbound AI-driven voice calls for automated customer support and appointment handling. Developed multi-gateway payment integration supporting Stripe, and Razorpay for seamless transactions within conversational systems. Deployed production-ready architecture to ensure scalability, high availability, and reliable load handling..

Conversational AI Systems & Voice Assistants: Designed and deployed multi-channel AI chat and voice assistants using Azure OpenAI GPT models, Qdrant vector DB, Node.js, and n8n workflows. Enabled clients to automate property queries and appointment bookings, achieving sub-1-second response times for voice assistants.

AI automation SaaS Platform: Put into production a SaaS automation platform for different industries with customizable chat assistants for website widgets and WhatsApp, enabling lead capture and business growth. Built workflows, analytics dashboards, and secure user/admin authentication using Next.js and Node.js.

E-commerce Chatbot using WooCommerce & MCP Server: Integrated WooCommerce stores with MCP Server and created an e-commerce chatbot that supports product search, order placement, status checks, and simple payment methods. Built APIs between the storefront and chatbot, implemented session management and cart workflows. Testing showed an improvement of about 55–70% in automated order handling, significantly reducing manual lookup time for agents.

Production Deployment & Testing: Took part in client meetings, requirement refinement, and production rollouts. Helped debug integration issues, monitored uptime, and collected post-deployment feedback to iterate on flows. Performed end-to-end testing of chatbot workflows and AI integrations to ensure reliability and smooth user experience. Estimated OpenAI API token usage and costs for different client scenarios, optimized prompts to reduce expenses by up to 20–25%, and contributed to pricing models for company website chatbot deployments.

Managed SEO and SEM activities to improve website visibility, search rankings, and paid campaign performance. Used Google Analytics and Microsoft Clarity to analyze traffic, user behavior, and optimize conversions. Performed page indexing and search performance monitoring through Google Search Console to ensure faster Google discovery.

EDUCATION

MSc. in Data Science (Merit Award)

Heriot-Watt University, Dubai Campus

GPA: 3.5

2024- 2025
School of Mathematics and Computer Science

Bachelor of Technology in Electronics and Communication Engineering

APJ Abdul Kalam Technological University, Kerala, India

CGPA: 8.51

2019-2023
Vimal Jyothi Engineering College

PROJECTS

- **Dissertation (A grade) –University Student Mental Health prediction and intervention using ML:** [Link](#) Preprocessed and cleaned multi-region student mental health datasets (missing values, feature engineering, correlation analysis, scaling). Built and

fine-tuned 8 ML models (tree-based and ensemble methods) achieving 95% accuracy in predicting stress, anxiety, and depression. Deployed the best models in a Flask web app for real-time predictions and adaptive interventions.

- **Danube Company Sentiment Analysis:** [Link](#) Performed Reddit comment sentiment analysis using NLP by collecting and cleaning user-generated text data, including tokenization, stopword removal, and lemmatization. Applied TF-IDF vectorization for feature extraction and trained SVM and Logistic Regression classifiers to categorize sentiments.
- **Heart Disease Prediction using ML** [Link](#): Multiple machine learning techniques were applied to predict myocardial infarction risk using both clinical and ECG image data. K-Means clustering segmented patients into three distinct risk groups, while Random Forest achieved the highest accuracy on clinical data (92.7%). For ECG image analysis, CNNs demonstrated strong performance (84.49%), with data augmentation improving generalization than MLP.
- **Data Analysis Application** [Link](#): Python GUI for large-scale document interaction analytics (3M+ records) using Dask for distributed processing. Included charts for top readers, popular browsers, countries, and related documents.
- **Amazon Reviews Analysis** [Link](#): Performed extensive data handling and exploratory data analysis (EDA) on Amazon product reviews. Applied text preprocessing, normalization, and feature representation using vector space models. Built and trained models including LSTM, BERT, and Hugging Face transformers for sentiment classification and topic modeling. Conducted hyperparameter tuning and sequence modeling to improve prediction accuracy and extract meaningful insights from customer reviews.
- **Kaggle Binary Classification Challenge**: Worked on a Bank Marketing dataset, performed feature engineering, and used ensemble models including LightGBM, XGBoost, and hybrid LightGBM + Neural Network models, achieving a score of 0.96971 (top score: 0.97801).

SKILLS

Programming: Python, JavaScript, C, C#

RAG & LLM Systems: LangChain.js, Hugging Face Transformers, Prompt Engineering, Fine-Tuning, AI Agents, Multi-Agent Workflows, Context Window Management, Token Budgeting, RAG Pipelines, Retrieval Optimization, Hallucination Mitigation

Machine Learning & Deep Learning: Supervised & Unsupervised Learning, Classification, Regression, Clustering, Feature Engineering, Model Optimization, Hyperparameter Tuning, Cross-Validation, SMOTE, Correlation Analysis, Neural Networks, scikit-learn, TensorFlow, PyTorch

AI Systems, Automation & APIs: REST APIs, API Integration, AI Agent Pipelines, n8n Workflows, MCP Server, WAHA Connectors, Webhooks, SaaS Integrations, Workflow Automation

NLP & Text Analytics: LSTM, BERT, Transformers, Tokenization, Lemmatization, Vector space representation, word embeddings, Sentiment Analysis, Emotion Detection, Text Classification

Database: MySQL, MongoDB, PostgreSQL, Qdrant, Vector Databases, Snowflake

Visualization & Analysis Tools: Tableau, Plotly, Seaborn, Matplotlib, D3.js, Google Analytics, Microsoft Clarity, Google Search Console

Developer Tools & Deployment: Docker, Git, Postman, Jupyter Notebook, VS Code, PyCharm, Cursor, AWS, OpenAI Codex, Gemini, Google AI Studio, Claude Code

Soft Skills: Client Communication, Team Collaboration, Critical Analysis, Requirement Analysis, Problem Solving, Attention to Detail, Adaptability & Learning Agility

CERTIFICATES AND INTERNSHIPS

Build AI Agents and Chatbots with LangGraph	LinkedIn Learning	2025
Machine Learning with Python Foundations	LinkedIn Learning	July 2025
BCG Data Science Job Simulation	-The Forage	April 2025
SPARKS Foundation	-Data Science & Business Analytics Internship (online)	2024(1 month)
Diploma in Python with React and Data Science	-Internship (STED Council Certification)	2023-24 (6 months)

ACHIEVEMENTS AND PARTICIPATION

GITEX GLOBAL 2025 (Dubai WTC): Attended GITEX 2025, explored AI technologies, met industry professionals, and gained insights into enterprise AI integration.

IELTS :Overall: 7.5

Awarded **UAE Golden Visa** for academic excellence

Volunteer & Member of Certification & Publishing Team, TANTRA Technical Fest

Bronze Medal- All Kerala Sahodaya Scholastic Achievement Test; Participated in All Kerala Sahodaya Inter-School competitions for Chess, Badminton, and Pencil Drawing.

PERSONAL DETAILS

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Nationality: Indian

Location: Jumeirah Lake Towers, Dubai

VISA Status: Golden Visa